Taha Fawzy Anwar Elshrif

Graduated with excellent with an honor degree

tahaelshrif1@gmail.com
+20 1551988337
https://tahafawzyelshrif.github.io/dev.taha/

Professional Summary

Software Engineer specializing in AI systems and full-stack development, with a strong focus on building practical, production-ready solutions. Experienced in integrating LLMs, automation, and scalable service architectures, from API design to deployment.

Experience

AI Intern – XClan (SEP 25 - JAN 26)

- Led a 5-member AI team for Care Now, an on-demand nurse platform, developing OCR, text standardization, and face verification solutions.
- Evaluated and selected models based on performance and cost; deployed AI services with FastAPI and planned scalable integration with other teams.
- Technologies: FastAPI, Siamese Networks, LLM APIs, Azure.

Cloud Computing Intern – SRTA-CITY (SEP 24 - JAN 25)

- Gained practical knowledge of cloud infrastructure, virtualization, and fault-tolerant system design.
- Explored ML applications in data centers for monitoring and trend analysis, including energy efficiency and edge computing.

Projects

- **Traffic Manager (Graduation Project)** Led a 10-member team in developing a smart traffic signal system to address congestion caused by static signal timings that do not adapt to current road conditions. Trained through simulation and integrated with an IoT prototype, achieving an average waiting time of 21 seconds per 30 minutes while maintaining fairness using Reinforcement Learning, Simulation of Urban Mobility (SUMO), Analysis of Variance (ANOVA), Computer Vision, FastAPI, and Flutter. LINK
- **CodeBuddy (Side Project)** Designed and implemented a full-stack AI learning platform for beginner developers, helping them overcome common programming blockers through structured answers based on their needs and device constraints. Designed a ReAct AI agent with RAG and tool integration, refined its handling , and manually tested LLM responses across 5 cases. Built the system with an HTML, CSS, Bootstrap, and JavaScript frontend and a scalable backend using Clean Architecture, FastAPI, PostgreSQL, Redis, queueing, encryption, deployment, and monitoring. LINK
- **AidMate (Side Project - Open Source)** Designed and developed an AI-powered first-aid assistant for providing initial guidance in urgent situations. Built an offline-capable RAG pipeline using TF-IDF semantic search and designed the full-stack application with FastAPI and Flutter for Android and Windows, using Provider for state management, Flutter Markdown, responsive UI, and customizable themes. LINK

- **Xiaomi Social Analysis (Side Project)** Performed real-time social media analysis to identify current issues and common customer complaints about popular Xiaomi products, and developed a sentiment analysis model achieving 70% accuracy. Built the pipeline using Kafka, Reddit, Python, and GPT for data collection, analysis, and insight generation. LINK

- **Inventory Risk Replenishment Assistant** Vibe-coded inventory management system developed after studying the supply chain domain, using structured engineering prompts and AI-assisted development. Reviewed and modified AI-generated code to implement inventory risk logic and improve demand forecasting using a moving-average approach. LINK

Core Technical Skills

Programming Languages: Python, JavaScript, Java, Bash

AI & Machine Learning: LLMs, AI Agents, RAG, Generative AI, Machine Learning, Deep Learning (CNNs), Reinforcement Learning, Computer Vision (YOLO), Model Optimization, AI System Design, Model Serving

Backend APIs: FastAPI, Flask, Node.js, REST APIs, API Design, Middleware, Rate Limiting, Stream Processing

Cloud Infrastructure: Docker, Azure, AWS, Terraform, Cloud Deployment, Fault Tolerance, Virtualization, Cloudflare

Software Engineering: Clean Code, Design Patterns, Client-Server Systems, System Architecture, Documentation, Agile Development, Testing

Databases: SQL, NoSQL, PostgreSQL, Redis, Firebase

Tools Platforms: Git, GitHub, Apache Kafka, n8n, MS Office

Data Engineering: ETL, Feature Engineering, Data Science Fundamentals

IoT: ESP32

Education

B.Sc. in Data Science
Faculty of Computers and Data Science, Alexandria University – 2025
GPA: 3.79 / 4 (Excellent with Honor Degree)

Certifications

IBM Badges (LLMs, Deep Learning & Generative AI, RL, AI Agents), Linux Foundation, Meta Advanced Databases.

Languages

Arabic: Native - English: B2